

Cost Analysis

2026-08-03

Contents

1	Assumptions	1
2	Executive Summary	1
3	Breakdown by grades	10
3.1	3200115	10
3.2	6010100	14
3.3	6010120	18

1 Assumptions

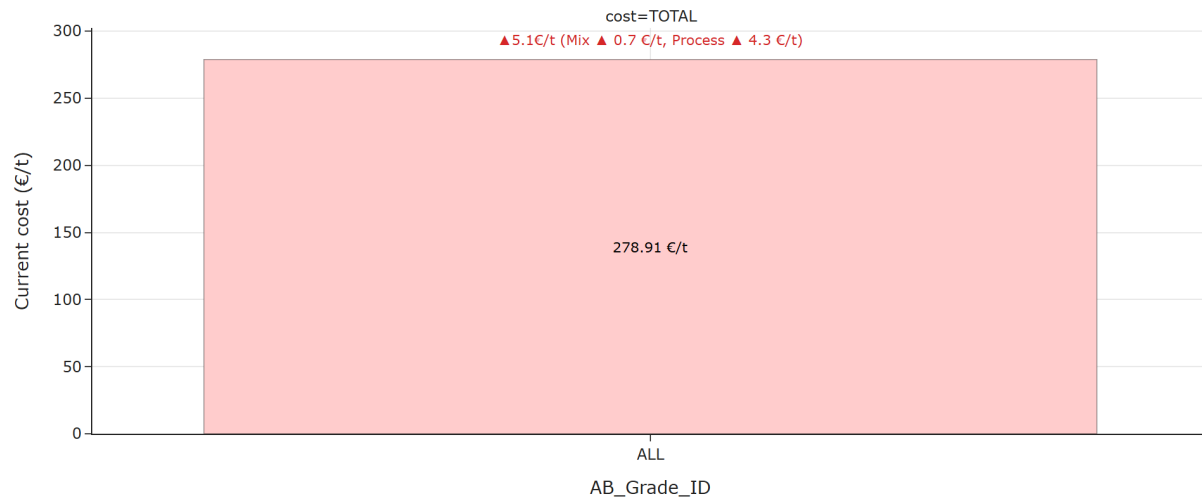
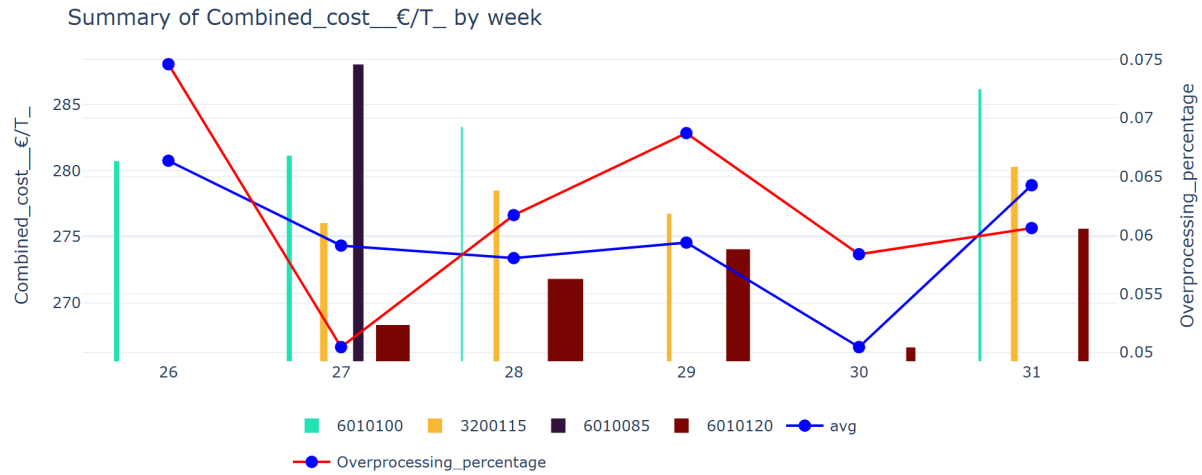
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** June 28, 2026 – August 03, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

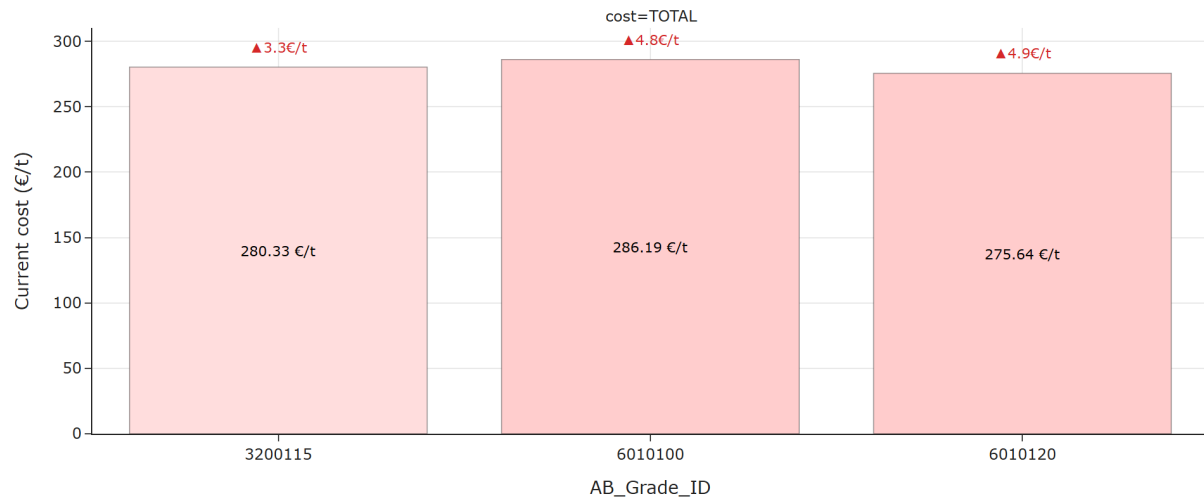
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw an increase of 5.06 €/t, resulting in 278.91 €/t.

Breaking down the overall cost change, +0.73 €/t was driven by a more expensive mix of grades, and +4.34 €/t was due to a deterioration in the process.



For grade 3200115, TOTAL cost saw an increase of 3.27 €/t, resulting in 280.33 €/t.

For grade 6010100, TOTAL cost saw an increase of 4.84 €/t, resulting in 286.19 €/t.

For grade 6010120, TOTAL cost saw an increase of 4.90 €/t, resulting in 275.64 €/t.

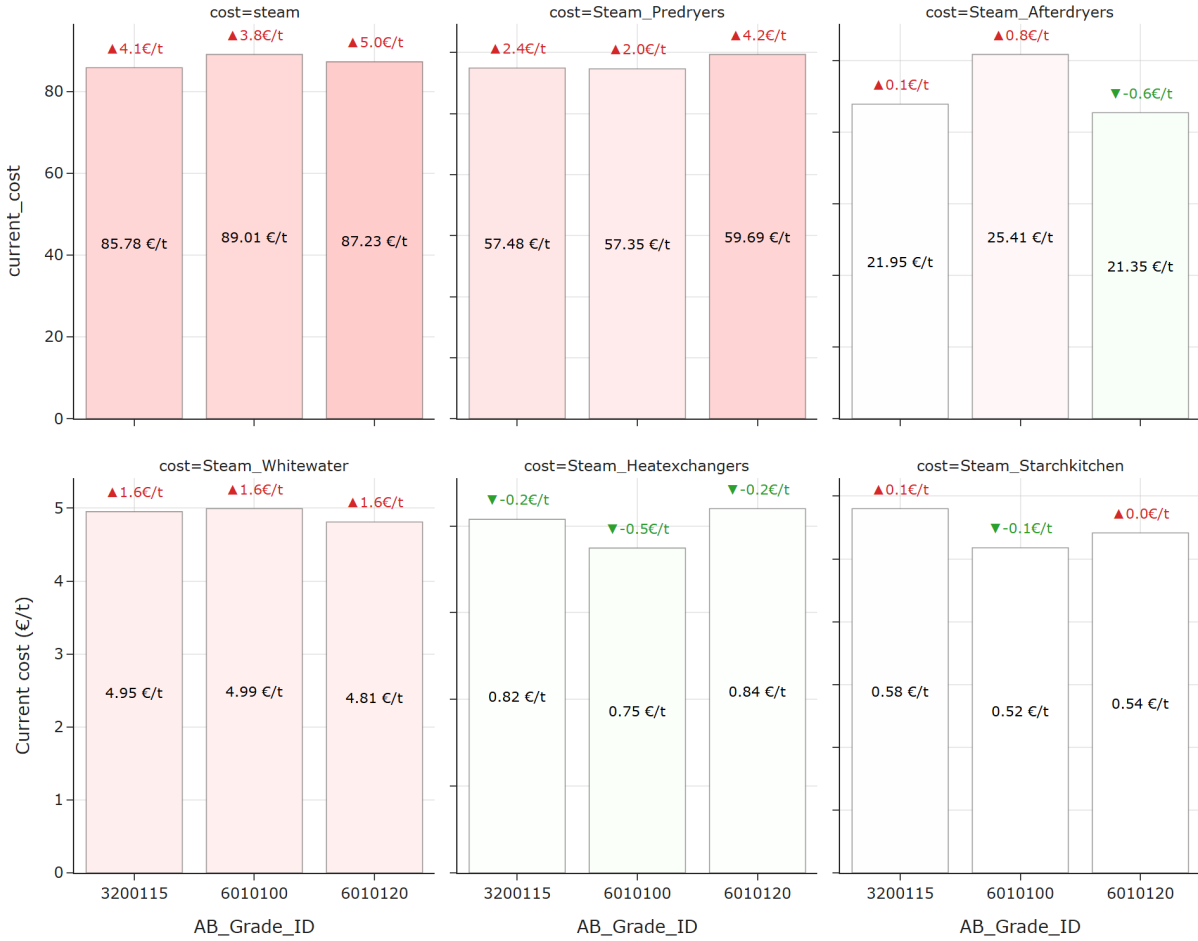
Looking across individual grades, the strongest cost increase was observed for grade 6010120 (+4.90 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam_€/T** increased on average (+4.30 €/t).
- **Electricity_€/T** decreased on average (-0.47 €/t).
- **Starch_€/T** increased on average (+0.29 €/t).
- **Fibre_cost_€/T** increased on average (+0.21 €/t).
- **Chemicals_€/T** increased on average (+0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_€/T** (+4.30 €/t on average), with the highest increase in grade 6010120 (+5.02 €/t) and the largest average decrease comes from **Electricity_€/T** (-0.47 €/t on average), with the biggest drop in grade 6010120 (-0.86 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** increased on average (+2.89 €/t).
- **Steam_Whitewater** increased on average (+1.61 €/t).
- **Steam_Heatexchangers** decreased on average (-0.30 €/t).
- **Steam_Afterdryers** increased on average (+0.10 €/t).
- **Steam_Starchkitchen** increased on average (+0.01 €/t).

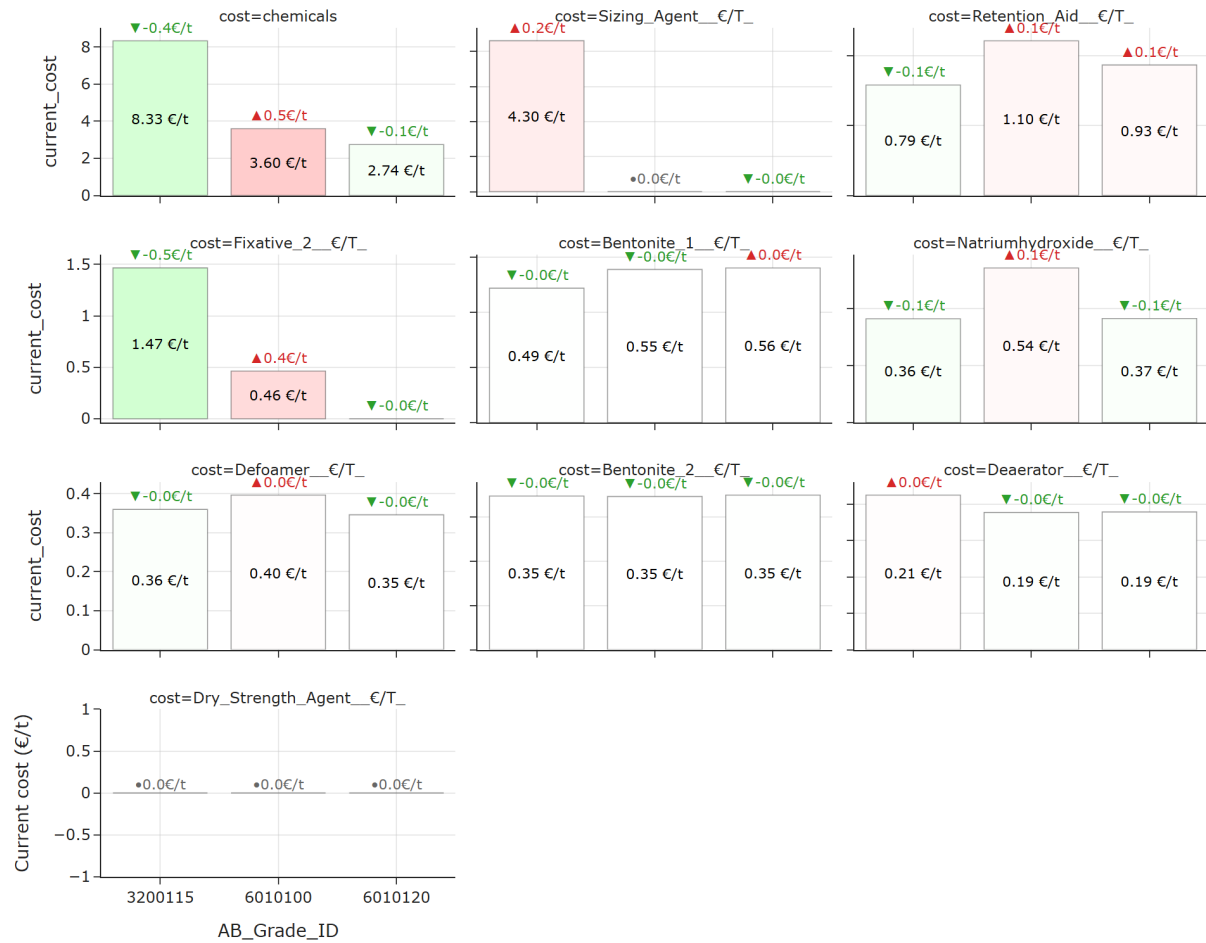
A more detailed breakdown by component shows that the largest average increase comes from **Steam_Predryers** (+2.89 €/t on average), with the highest increase in grade 6010120 (+4.23 €/t) and the largest average decrease comes from **Steam_Heatexchangers** (-0.30 €/t on average), with the biggest drop in grade 6010100 (-0.53 €/t). These components stand out and may require further investigation.



Average change by electricity:

- **Electricity_Other** decreased on average (-0.73 €/t).
- **Electricity_Wiresection** increased on average (+0.40 €/t).
- **Electricity_Specific** increased on average (+0.20 €/t).
- **Electricity_Freqconverters** decreased on average (-0.17 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.11 €/t).
- **Electricity_Predryer** decreased on average (-0.07 €/t).
- **Electricity_Afterdryer** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Wiresection** (+0.40 €/t on average), with the highest increase in grade 6010100 (+0.46 €/t) and the largest average decrease comes from **Electricity_Other** (-0.73 €/t on average), with the biggest drop in grade 3200115 (-0.87 €/t). These components stand out and may require further investigation.

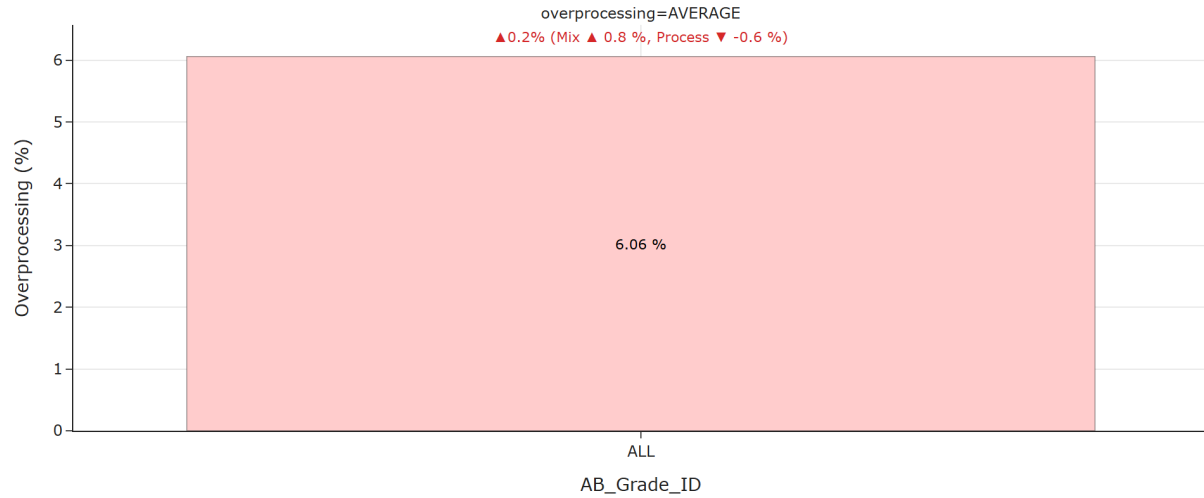


Average change by chemicals:

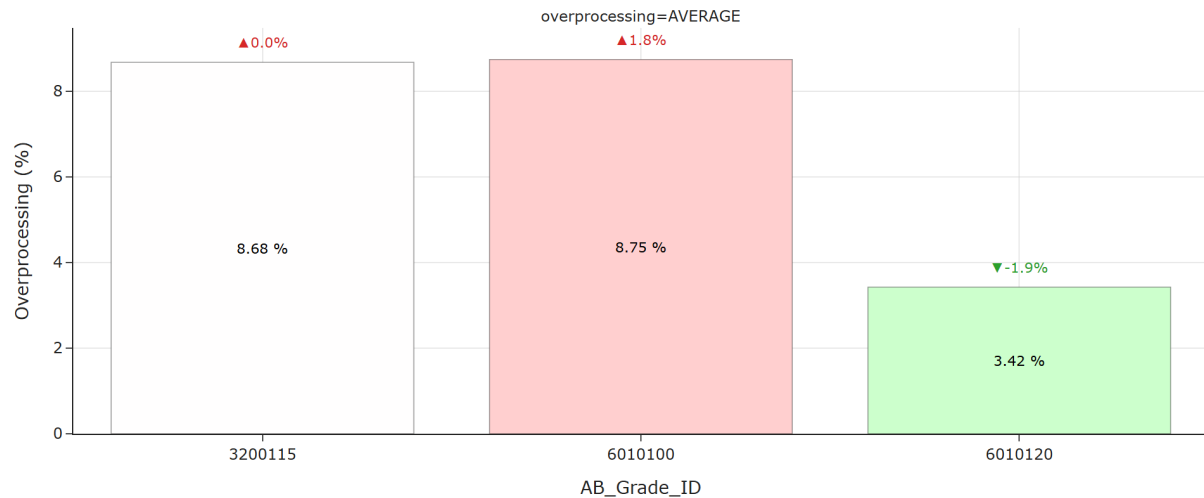
- **Sizing_Agent_€/T** increased on average (+0.18 €/t).
- **Fixative_2_€/T** decreased on average (-0.04 €/t).
- **Retention_Aid_€/T** increased on average (+0.03 €/t).
- **Natriumhydroxide_€/T** decreased on average (-0.02 €/t).
- **Defoamer_€/T** decreased on average (-0.01 €/t).
- **Bentonite_2_€/T** decreased on average (-0.00 €/t).
- **Deaerator_€/T** decreased on average (-0.00 €/t).
- **Bentonite_1_€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.18 €/t on average), with the highest increase in grade 3200115 (+0.18

€/t) and the largest average decrease comes from **Fixative_2_€/T** (-0.04 €/t on average), with the biggest drop in grade 3200115 (-0.45 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 0.17 %, resulting in 6.06 %.



For grade 3200115, TOTAL overprocessing saw an increase of 0.05 %, resulting in 8.68 %.

For grade 6010100, TOTAL overprocessing saw an increase of 1.80 %, resulting in 8.75 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 1.90 %, resulting in 3.42 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010100 (+1.80 %) and the strongest overprocessing decrease was observed for grade 6010120 (-1.90 %). These movements stand out and may require further investigation.



Average change by component:

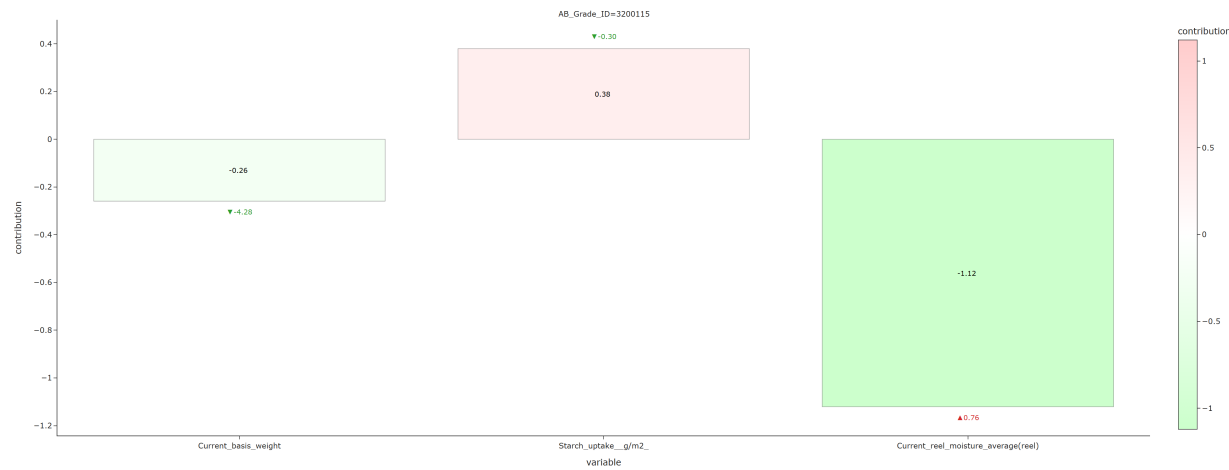
- **Burst** increased on average (+1.72 %).
- **SCT_CD** decreased on average (-0.70 %).
- **CMT30** increased on average (+0.14 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+1.72 % on average), with the highest increase in grade 3200115 (+1.72 %) and the largest average decrease comes from **SCT_CD** (-0.70 % on average), with the biggest drop in grade 6010120 (-2.18 %). These components stand out and may require further investigation.

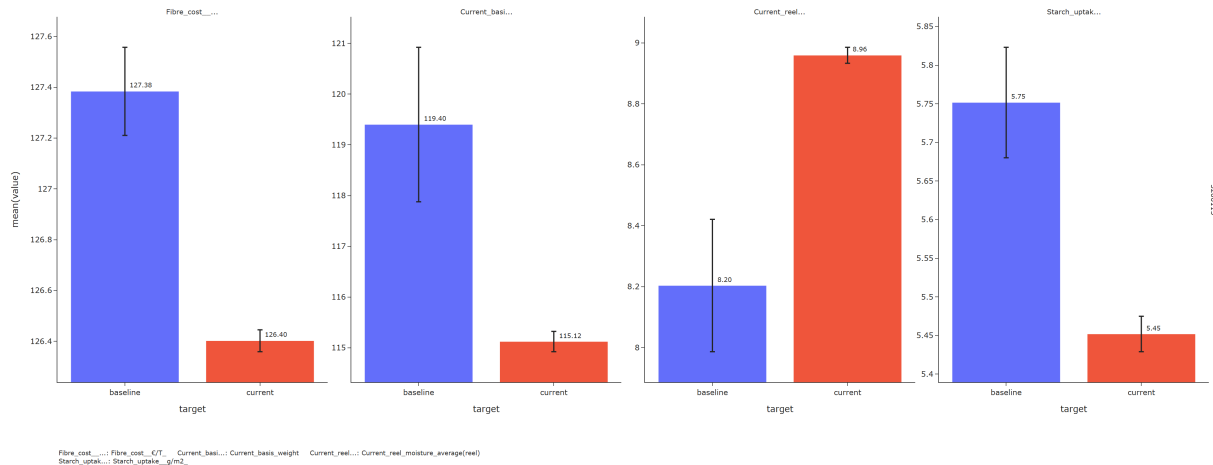
3 Breakdown by grades

3.1 3200115

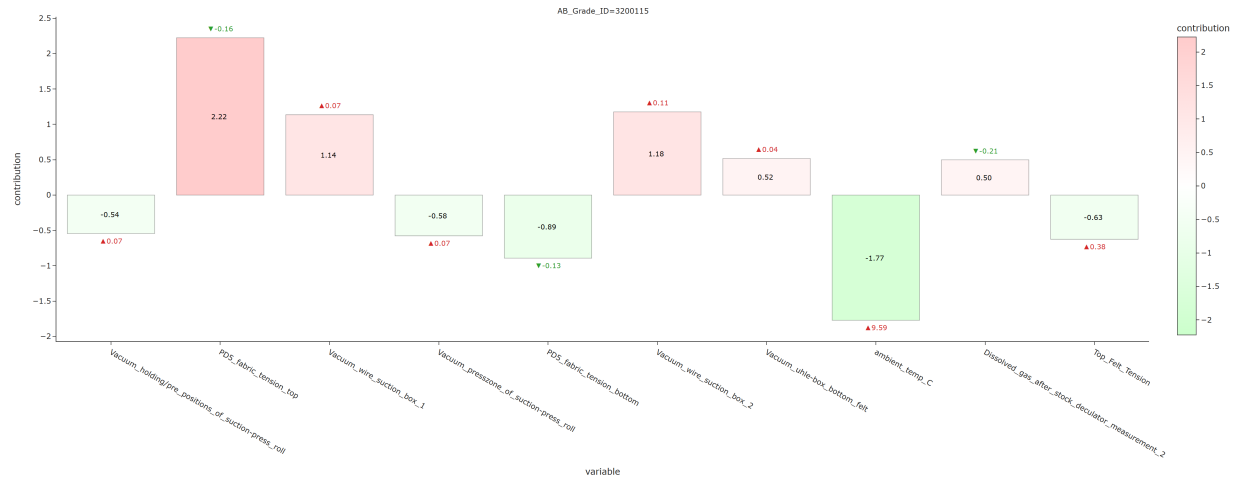
3.1.1 Fiber cost



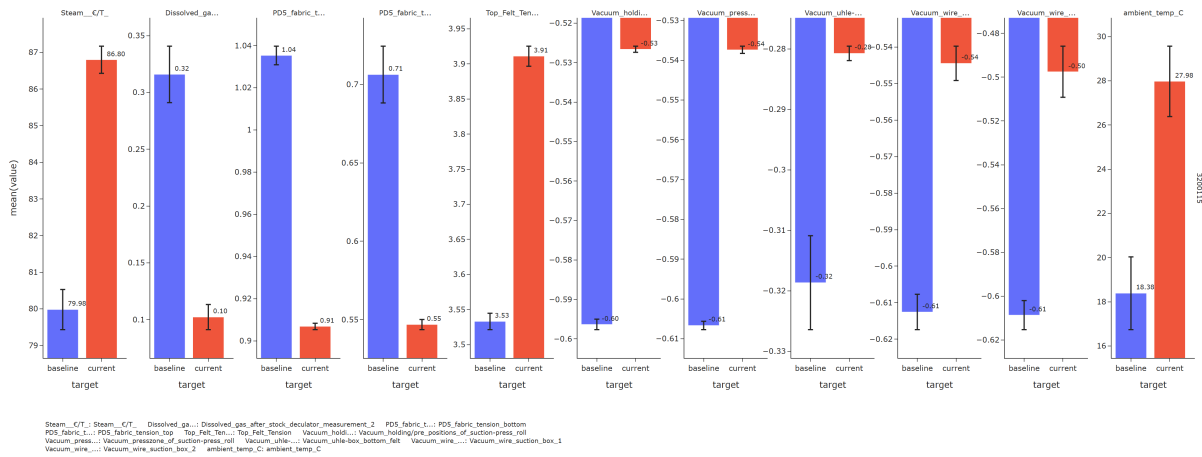
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Current reel moisture average(reel) increased (+0.76)



3.1.2 Steam cost



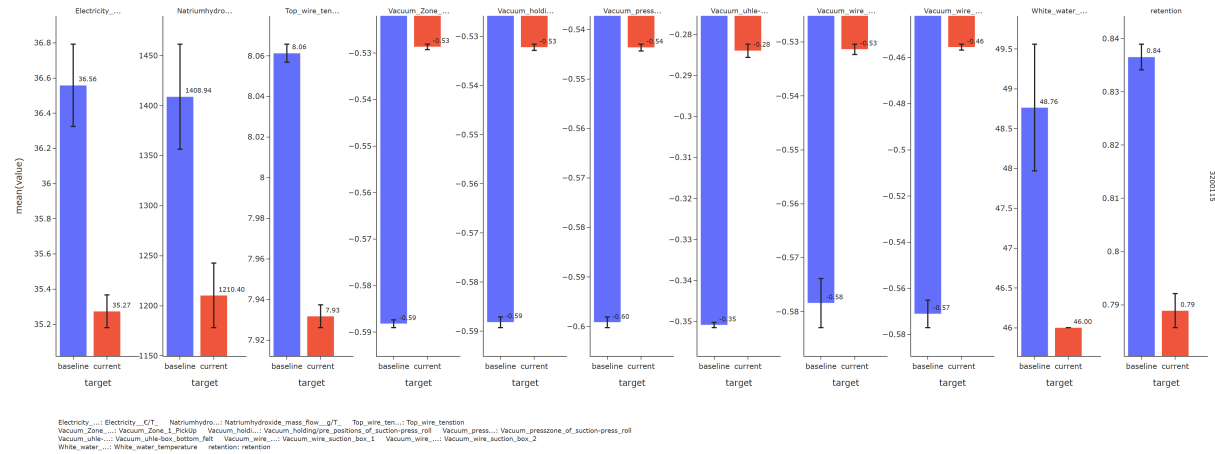
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Ambient temp C increased (+9.59) - PD5 fabric tension bottom decreased (-0.13) - Top Felt Tension increased (+0.38) - Vacuum presszone of suction-press roll increased (+0.07) **Cost-increasing drivers:** - PD5 fabric tension top decreased (-0.16) - Vacuum wire suction box 2 increased (+0.11) - Vacuum wire suction box 1 increased (+0.07)



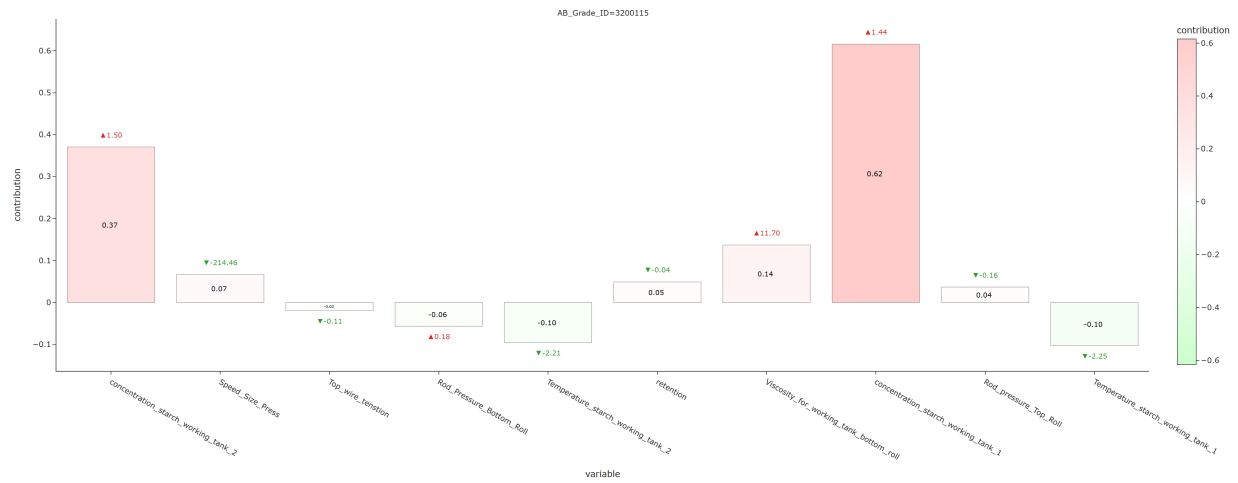
3.1.3 Electricity cost



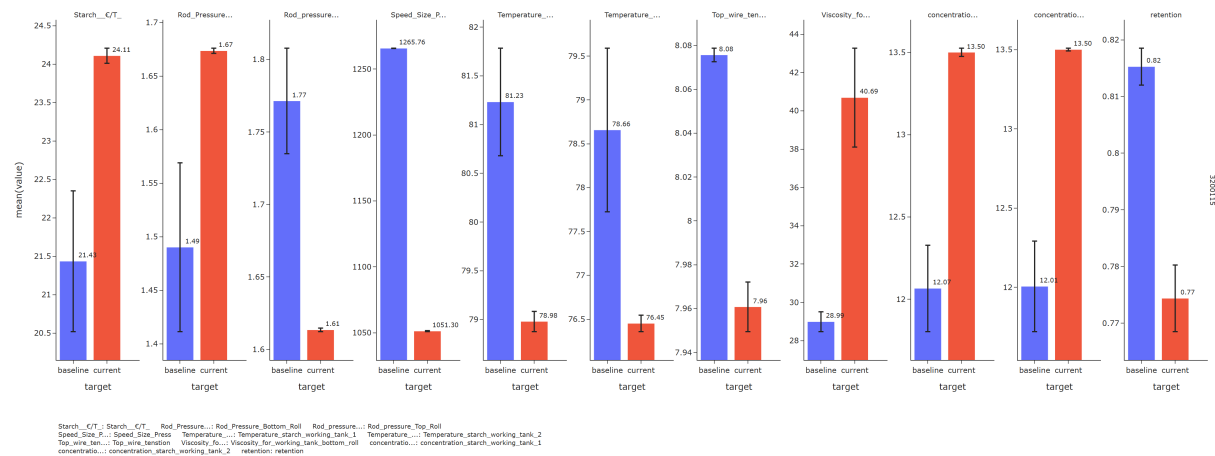
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Top wire tension decreased (-0.13) - Vacuum wire suction box 1 increased (+0.05) - Vacuum wire suction box 2 increased (+0.12)



3.1.4 Starch cost

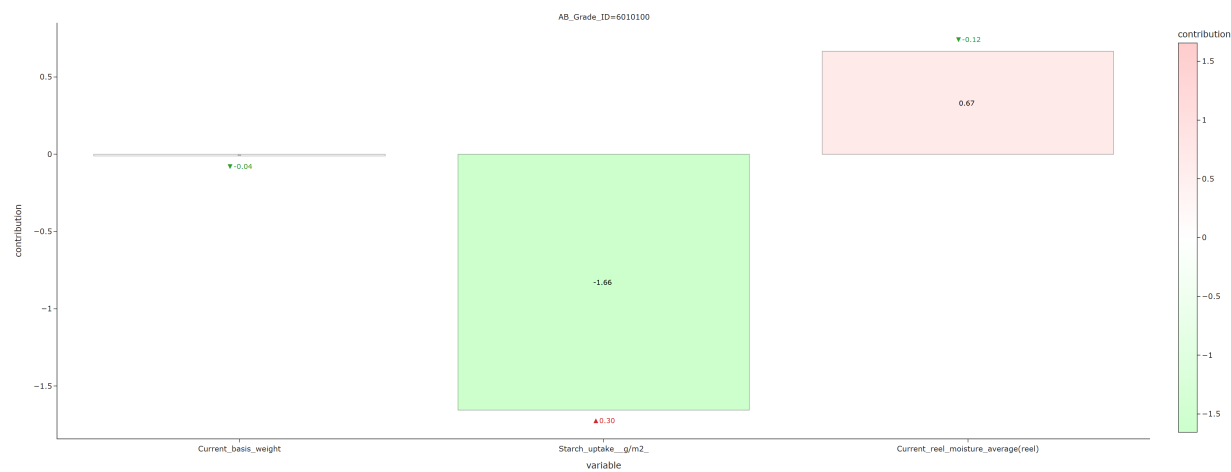


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Concentration starch working tank 1 increased (+1.44) - Concentration starch working tank 2 increased (+1.50) - Viscosity for working tank bottom roll increased (+11.70)

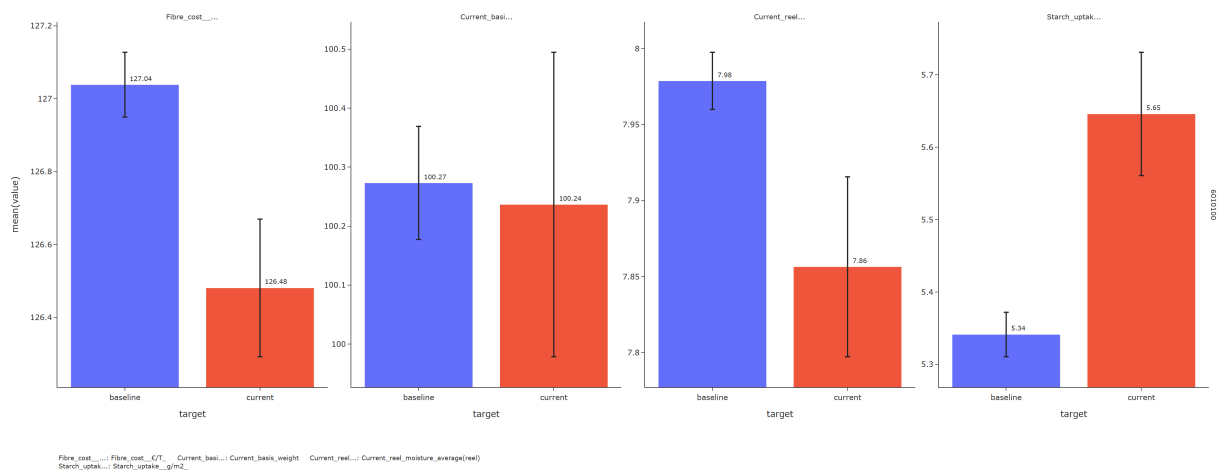


3.2 6010100

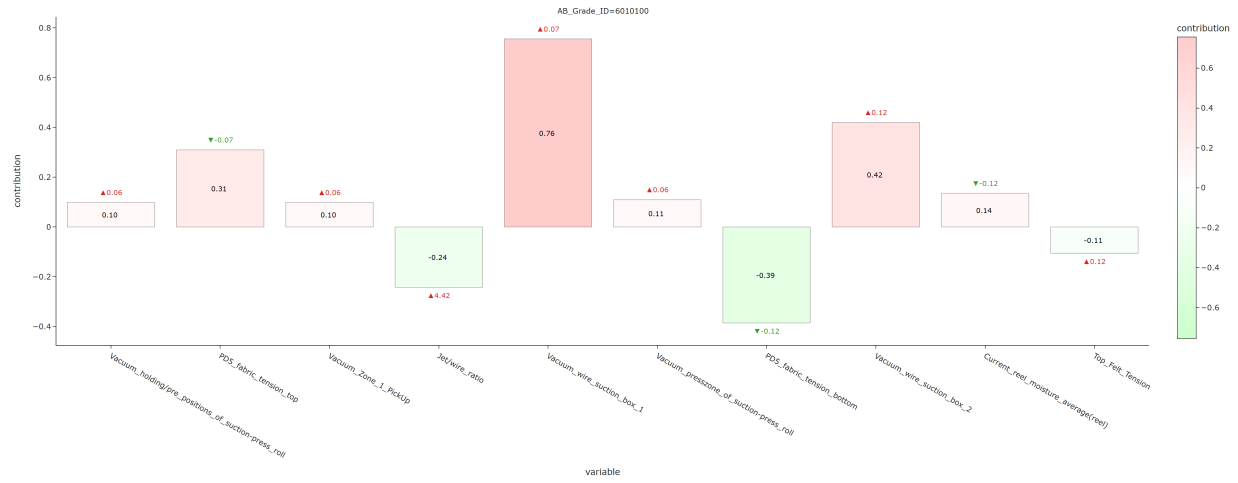
3.2.1 Fiber cost



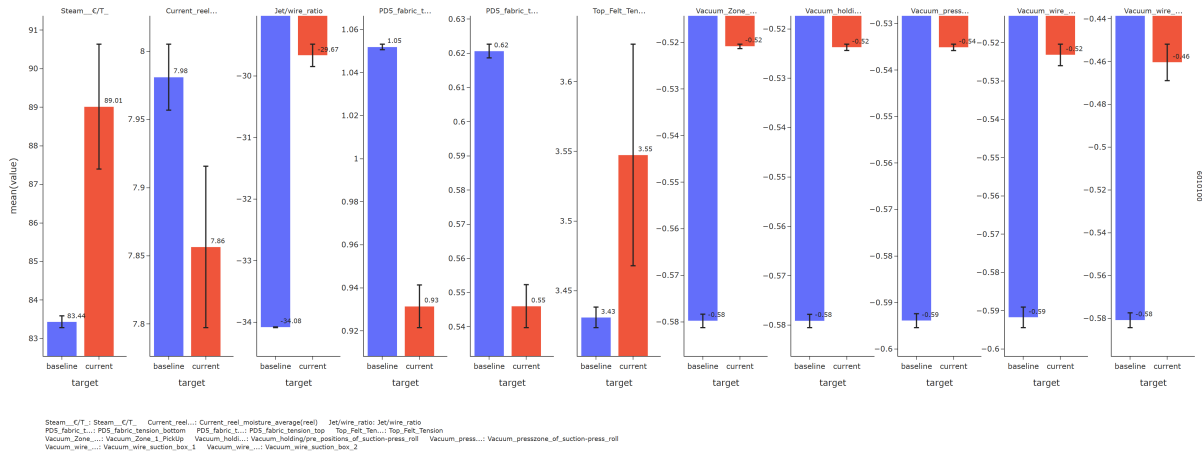
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Starch uptake g/m2 increased (+0.30)



3.2.2 Steam cost



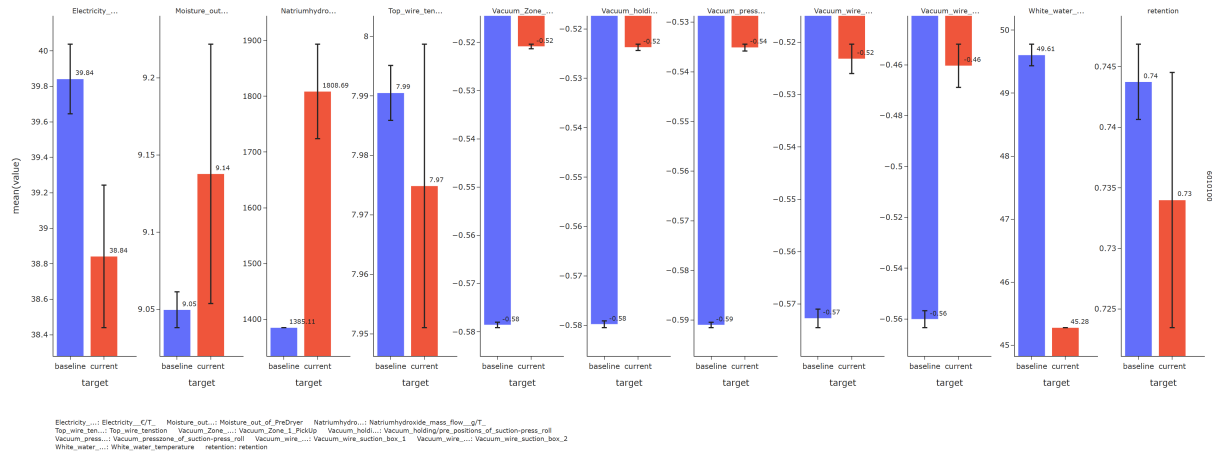
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - PD5 fabric tension bottom decreased (-0.12) - Jet/wire ratio increased (+4.42) **Cost-increasing drivers:** - Vacuum wire suction box 1 increased (+0.07) - Vacuum wire suction box 2 increased (+0.12) - PD5 fabric tension top decreased (-0.07)



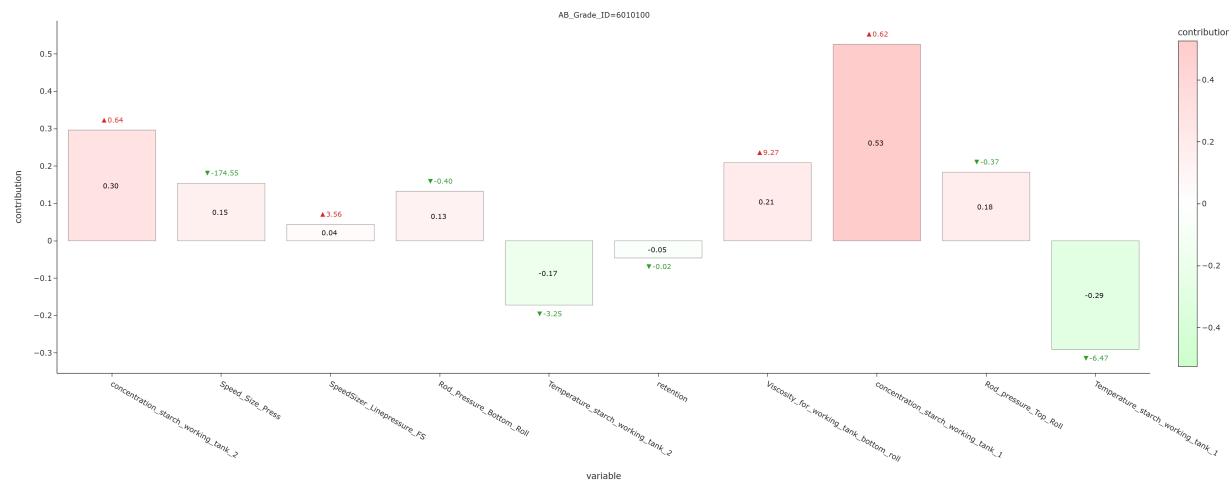
3.2.3 Electricity cost



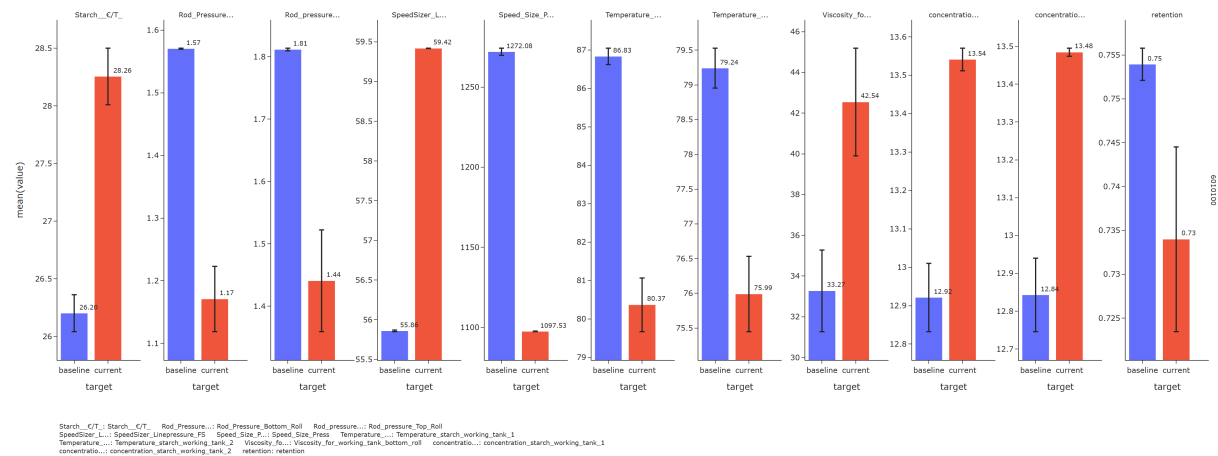
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Retention decreased (-0.01) - White water temperature decreased (-4.33) - Vacuum wire suction box 1 increased (+0.05)



3.2.4 Starch cost

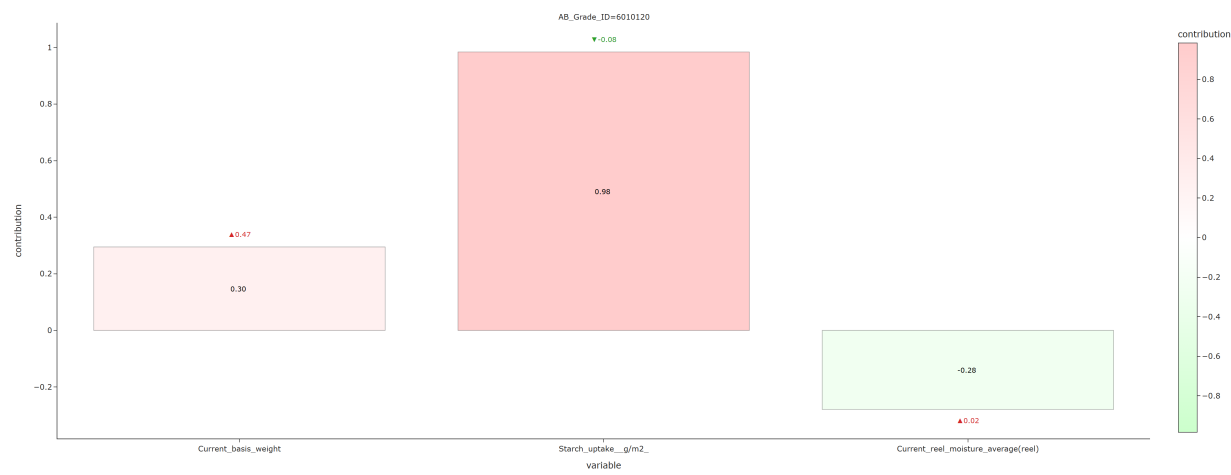


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Temperature starch working tank 1 decreased (-6.47) **Cost-increasing drivers:** - Concentration starch working tank 1 increased (+0.62) - Concentration starch working tank 2 increased (+0.64)

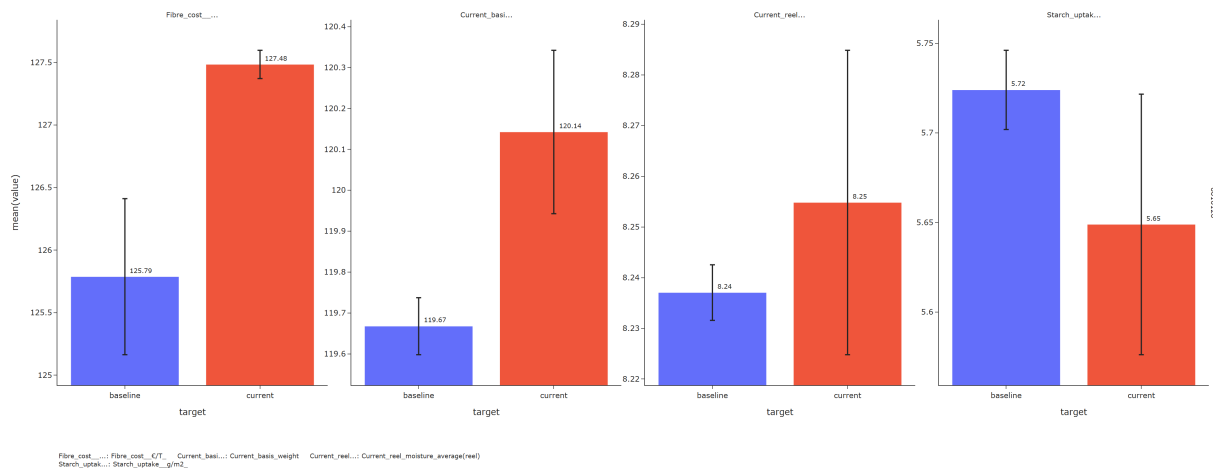


3.3 6010120

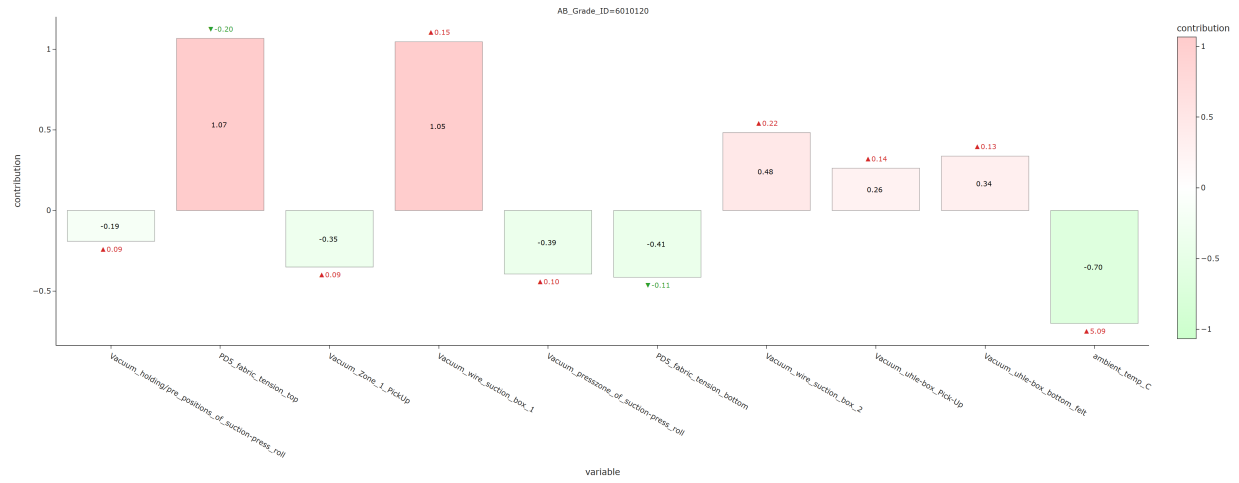
3.3.1 Fiber cost



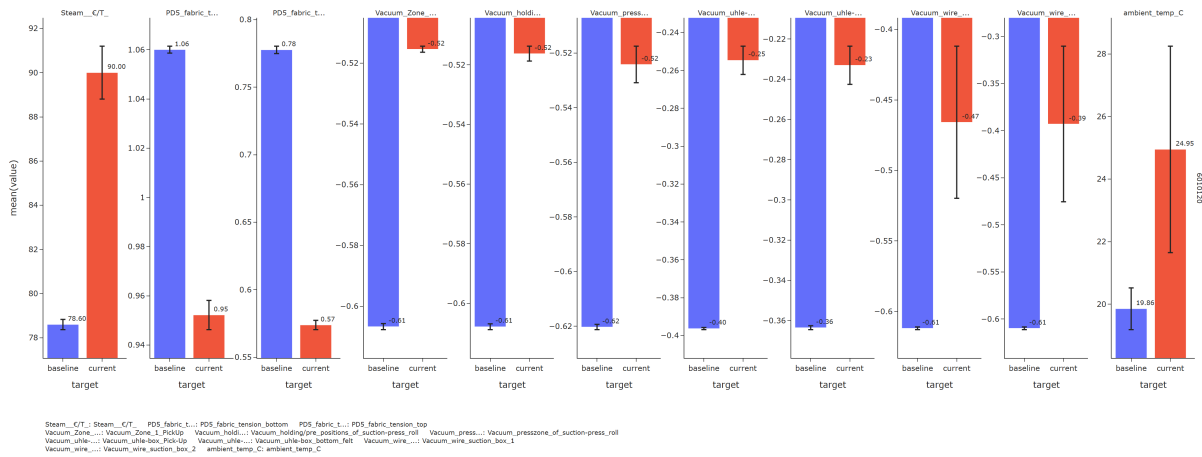
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Starch uptake g/m2 decreased (-0.08)



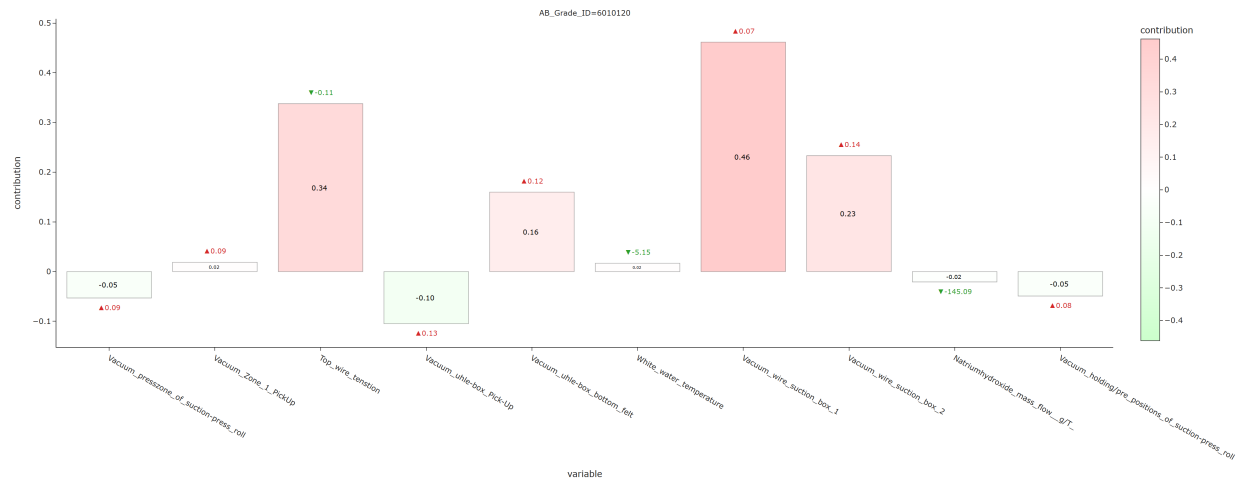
3.3.2 Steam cost



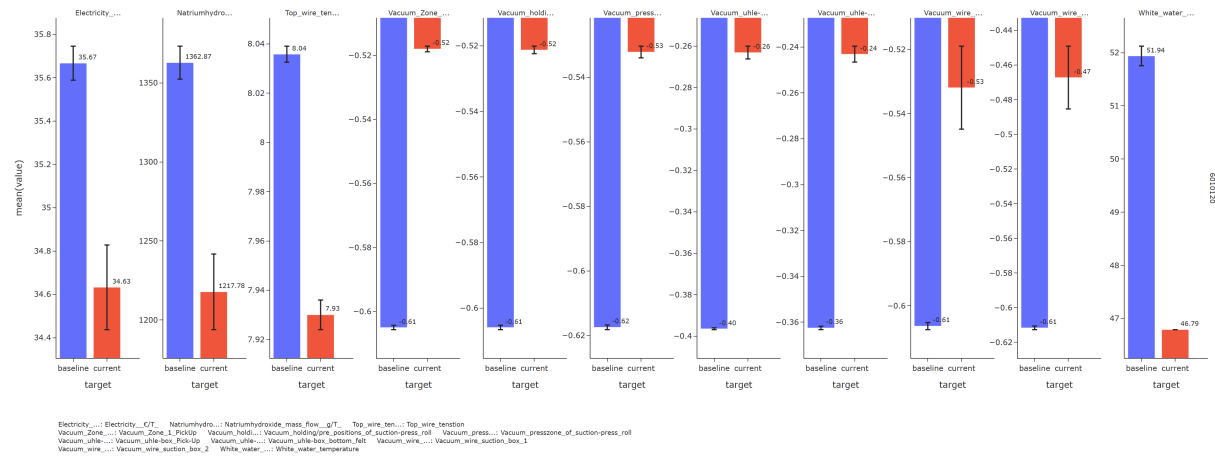
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Ambient temp C increased (+5.09) - PD5 fabric tension bottom decreased (-0.11) - Vacuum presszone of suction-press roll increased (+0.10) **Cost-increasing drivers:** - PD5 fabric tension top decreased (-0.20) - Vacuum wire suction box 1 increased (+0.15) - Vacuum wire suction box 2 increased (+0.22)



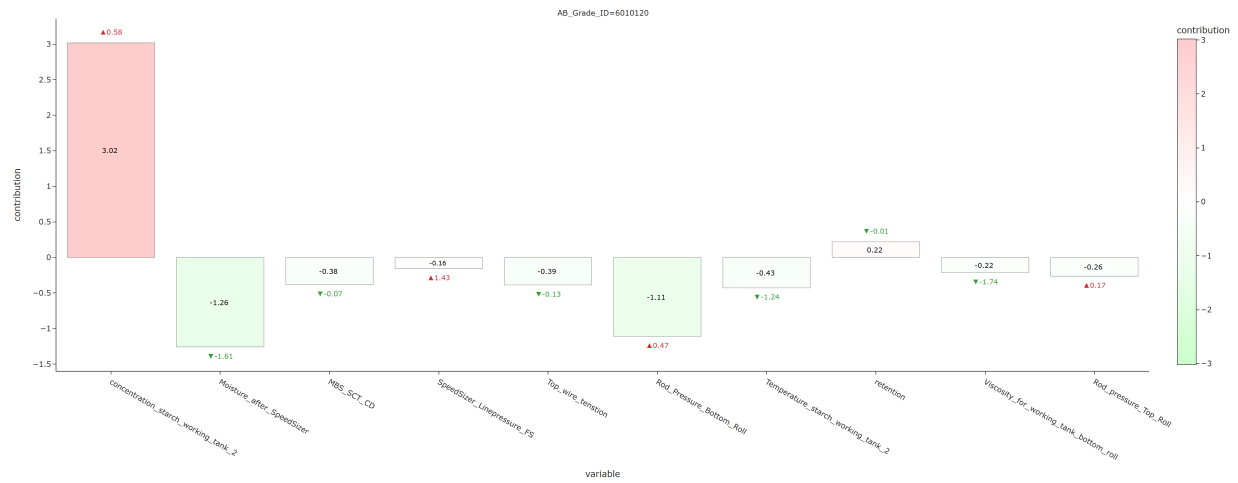
3.3.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Vacuum wire suction box 1 increased (+0.07) - Top wire tension decreased (-0.11)



3.3.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Moisture after SpeedSizer decreased (-1.61) - Rod Pressure Bottom Roll increased (+0.47) **Cost-increasing drivers:** - Concentration starch working tank 2 increased (+0.58)

